Explorative Data Analysis Report

1 Introduction

Rektum Cancer dataset involves MRIs which all gathered from Medipol Mega University Hospital. Machine model and imaging configurations varies imaging to imaging, also each patient has 2 types of imagings: ADC and T2, which we call ***modality***. Thats why a deep explorative analysis is mandatory. In this report, you will be finding from which perspectives data will be analyzed and results of each analyze. In addition, a conclusion -a general overview- on data and **preprocessing steps needs to be followed later on** will be provided.

1.1 Dataset Size and Format

Dataset consists of varying number of batches and each batch followes a strict folder hierarchy as below:

```
batch-1/
 ├── NONRESPONDER -OR- RESPONDER -OR- INTERMEDIATE/
 |   ├── [patient-name]/
 |   |   ├── ADC/
 |   |   |   ├──imaging-file-name.nrrd
 |   |   |   └──segmentation-file-name.nrrd
 |   |   ├── T2/
 |   |   |   ├──imaging-file-name.nrrd
 |   |   |   └──segmentation-file-name.nrrd

 ...
```

When all the data is counted and translated into a table; modality, label and number of each category distribution seems like below:

Table 1
Distribution of response groups by modality

Modality	Intermediate	Non-Responder	Responder	Total
ADC	52	44	92	188
T2	52	44	92	188
Total	**104**	**88**	**184**	**376**

Total of size of 188 patients makes the dataset considerably a small sized dataset, a support might be necessary. Having 2 different modalities is an adventage, however, needs to be processeed carefully to utilize the data richness properly. Additionallyi data has a significant class-imbalance. So this makes:

- using **class-weighted loss a considerable idea**,
- using **statified train-test seperation a mandatory approach**. **[!]**

1.2 Manifest

To analyze the dataset from a single center, a *manifest.csv* is created using the script *manifest_builder.py*. The distribution in [1.1] is extracted with the help of *manifest.csv*.

Manifest has following columns:

- `sample_id`

- batch_id
- response_group
- patient_name
- patient_id
- modality
- patient_path
- modality_path
- nesting_depth
- image_path
- mask_path
- image_filename
- mask_filename
- nrrd_file_count
- other_file_count
- image_candidate_count
- mask_candidate_count
- image_dimension
- image_shape_x
- image_shape_y
- image_shape_z
- image_dtype
- image_encoding
- image_spacing_x_mm
- image_spacing_y_mm
- image_spacing_z_mm
- image_origin
- mask_dimension
- mask_shape_x
- mask_shape_y
- mask_shape_z
- mask_dtype
- mask_encoding
- mask_spacing_x_mm
- mask_spacing_y_mm
- mask_spacing_z_mm
- mask_origin
- shape_matches
- spacing_matches
- origin_matches
- is_valid_sample
- validation_errors
- manifest_created_at

By using these metadatas, accessing the data is easier using the code and comparing spatial aspects such as *shape* and *origin* easier. Assessing the mask and imaging mismatches can be done easily and clearly.

All thoughout the exploratory data analysis, *manifest.csv* is used as the only interface to access the data. **However, it's crucial to update the *manifest.csv* when anew batch arrives, otherwise, analyze would lose it's relevance and validity**.

1.3 Next Steps

Each type of data needs to be analyzed -or processed- in it's own way. MRIs, for instance, are stack of slides of multiple 2D images, representing a 3D volume in real space. However, just like resampling images to one single dimension when feeding them to a CNN model, MRIs needs to be 'resampled' to a certain shape and spacing, in order to make sure all the imagings are representing the same volume in the real space. In order to find out all kinds of mismatches and fixing them, we need to apply following analyzes on the dataset:

1. **Null and Duplicate Check.** Must-do for all kinds of datasets, such anomalies harm the training and all kinds of statistics.
2. **Dimension Analyze (spacing, shape and origin).** Resolution in 2D images are equivelent to **shape** in MRIs. Spacing is unique to MRIs, represents **the real volume in millimeters of each voxel.** An imbalance in any of those needs to be fixed by **resampling and registering**, to prevent model memorize the shapes instead of patterns in the imaging.
3. **Volume Analyze.** Tumors' real world volume equivelences needs to be analyzed before training in order to find out abnormal big tumors, **an inconsistency in the spacing-shape translation of the imaging**, such a sample would harm the training. Also **if volume is already clearly an indicate of the target label** (for example big tumors are labeled non-reponder and smaller tumors are labeled responder to treatment) deep learning might be overkill and a simple XGBoost model might work better.
4. **Intensity (Brightness) Analyze.** Intensity imbalance confuses the model and makes it more complicated to learn from it. T2 imagings are normalized using **z-score normalization**, as a result, each imaging has a similar intensity distribution, preventing model to memorize the difference in intensities instead of patterns. However, **ADC modality should not be normalized** because each voxel represents a quantitative value instead of relative signal image.

2 Explorative Data Analysis

2.1 Nulls, Duplicates and Mask-Image Mismatches

Among all 188 patients, **no null or duplicate imaging exists**. A little mismatch over 15 patients is resolved and updated before, recently each of 188 patients has 2 modalities and 1 image-mask pair for each modality.

However, one patient has mask-image shape mismatch:

```
GUL, ABDULLAH
```

Unfortunately, this is not fixable or normalizable etc. straight up broken data. **We can't use this in the training. [1]**

2.2 Dimension Analysis

Each MRI sample has 3 dimensions: **(width, height, depth)** (order might differ due to library differences). In the following graphs; x, y, z will be equivelent to width, height, depth respectively. Distribution of these will give us idea about how big the shape imbalance and how big the memory necessity is. In the table below, you can find the statictical distribution of each dimension for T2 and ADC seperately:

Table 2
T2 image-shape statistics

Statistic	Image Shape X	Image Shape Y	Image Shape Z
Count	188.000000	188.000000	188.000000
Mean	445.521277	444.414894	31.218085
Std	133.118276	134.208148	6.873984
Min	256.000000	240.000000	15.000000
P25	352.000000	352.000000	25.000000
P50	400.000000	400.000000	30.000000
P75	480.000000	480.000000	36.000000
Max	768.000000	768.000000	65.000000

Table 3
ADC image-shape statistics

Statistic	Image Shape X	Image Shape Y	Image Shape Z
Count	188.000000	188.000000	188.000000
Mean	191.723404	216.808511	34.122340
Std	55.140929	44.434721	7.385557
Min	96.000000	96.000000	15.000000
P25	128.000000	188.000000	29.000000
P50	224.000000	224.000000	35.000000
P75	240.000000	256.000000	38.000000
Max	384.000000	328.000000	60.000000

In the figure below, you can find the graphical distribution of each dimension:

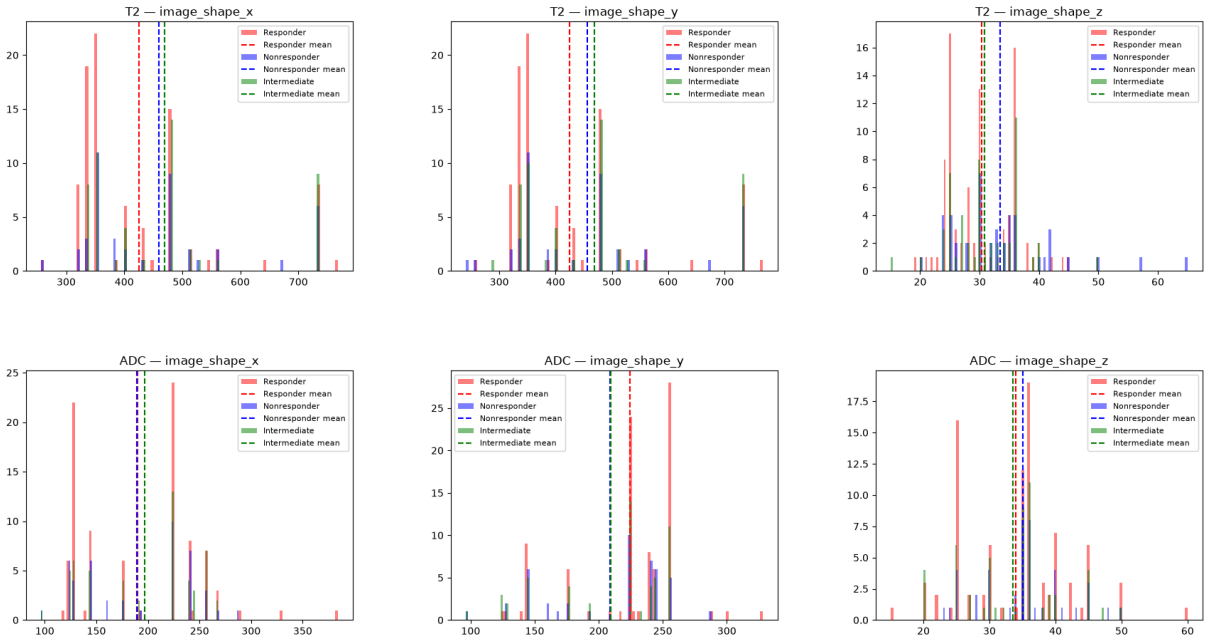


Figure 1: Dimensional distribution

We can conclude 2 points from the analyze:

1. Dimension distribution does not distinguish classes in either modalities.
2. Shape difference is very large, resampling all images to the same dimension might cause loss of information on large images and generation of too much artifical data on small images. **Instead, we can crop the tumor are only**, this would minimize the huge dimension difference and prevent model to focus on unnecessary large areas on the image. After cropping, resampling will be done. **[2]**

Other spatial information is **spacings**, a similar table and graph is given below:

Table 4
T2 image-spacing statistics

Statistic	Image Spacing X (mm)	Image Spacing Y (mm)	Image Spacing Z (mm)
Count	188.000000	188.000000	188.000000
Mean	0.533377	0.533377	4.114692
Std	0.229270	0.229270	1.419615
Min	0.271739	0.271739	2.999975
P25	0.357143	0.357143	3.000004
P50	0.397727	0.397727	3.500000
P75	0.625000	0.625000	4.125000
Max	1.171880	1.171880	9.227272

Table 5
ADC image-spacing statistics

Statistic	Image Spacing X (mm)	Image Spacing Y (mm)	Image Spacing Z (mm)
Count	188.000000	188.000000	188.000000
Mean	1.332626	1.332626	4.566809
Std	0.415194	0.415194	1.551235
Min	0.781250	0.781250	2.999979
P25	0.819672	0.819672	3.000000
P50	1.388889	1.388889	3.600014
P75	1.785714	1.785714	6.599999
Max	1.979167	1.979167	8.500000

In the graph below, you can find the distribution of each dimension's spacing:

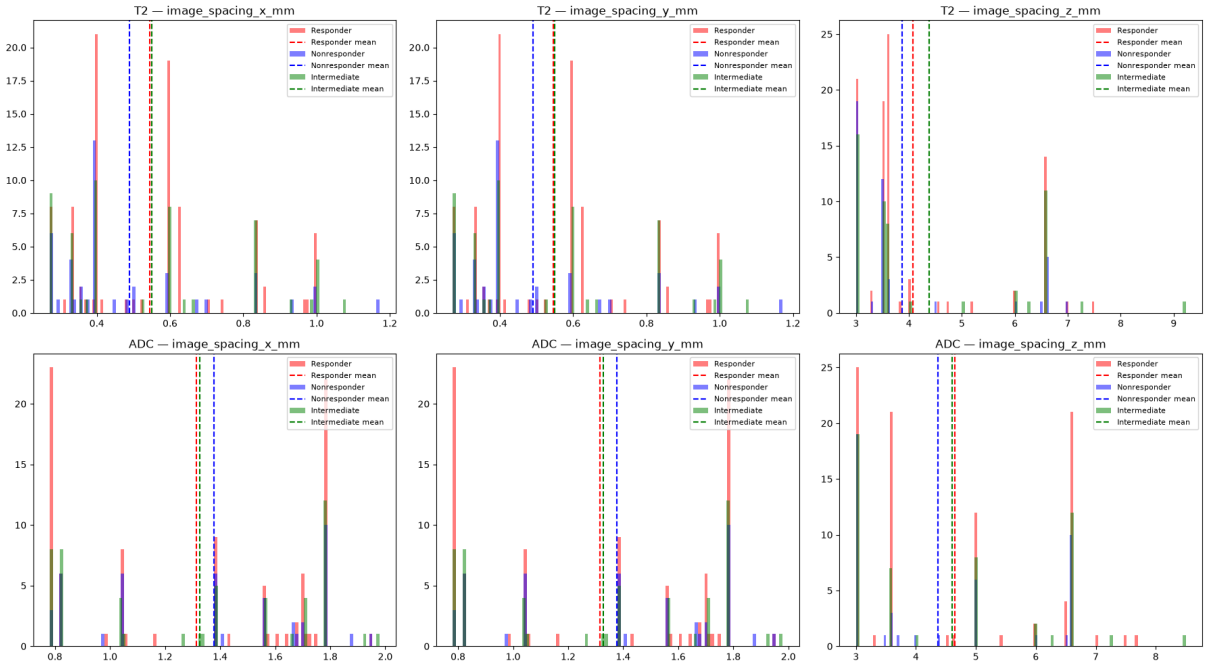


Figure 2: Spacing

2.3 Tumor Volume Analysis

If distinct classes are seperable on tumor volume, this would ease our work very much. In order to see this, we need to:

1. Count the number of tumor voxels in a mask,
2. Multiply the number of voxel by the mm^3 volume of each voxel in corresponding image. As a result, we will get the volume of the tumor on that one image. Then, we can have the distribution graphs.

You can find the related graphs below:

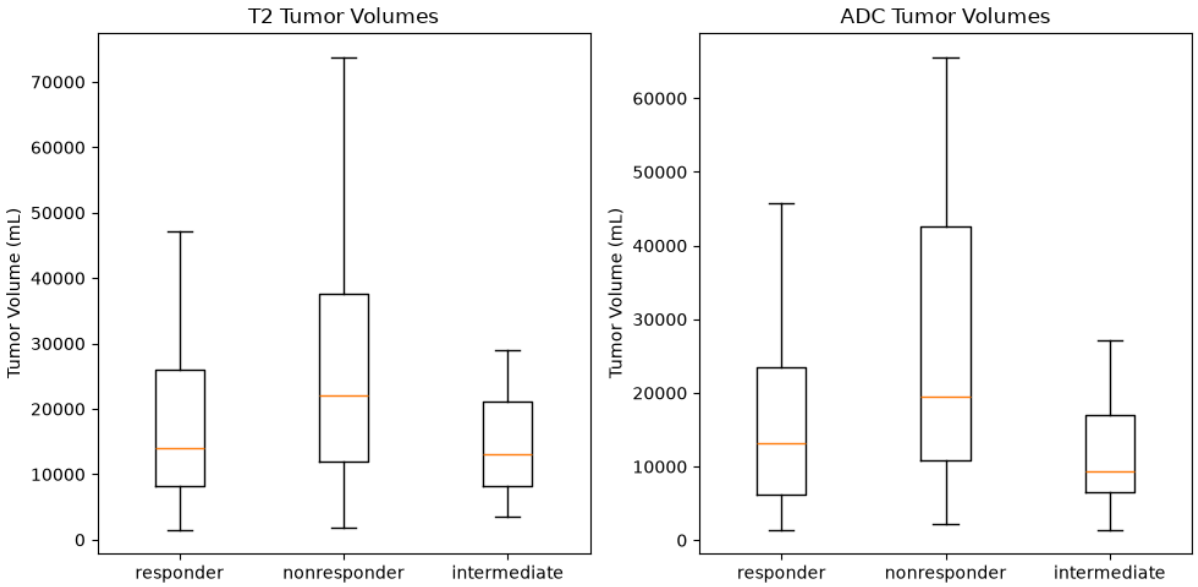


Figure 3: Tumor-volume boxplot

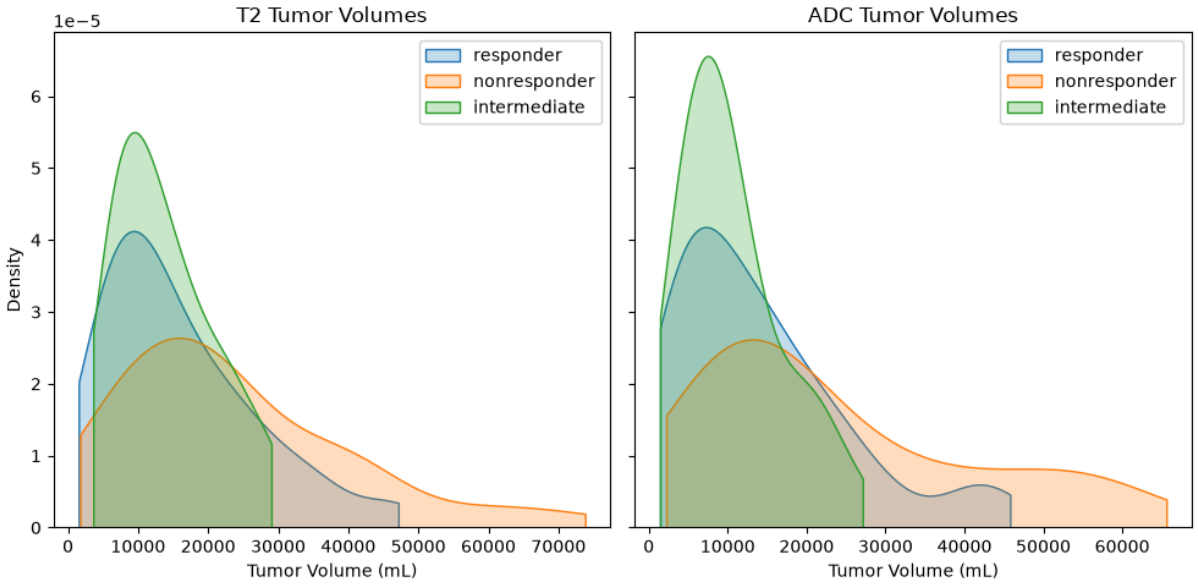


Figure 4: Tumor-volume density distribution

From the graphs, we can conclude that **tumor volumes does not seperate classes**, unfortunately. Each class have a similar distribution on volume but only number of samples belonging to each class differs.

2.4 Intensity Analysis

Intensity imbalance confuses the model. Counting the intensity of all voxels on an imaging and then calculating the mean and percentiles will help us to realize does imbalance exist, or how large it is. Below, the tables shows the intensity of each voxels in each image (for example mean of mean intensity of all images):

Table 6
All imaging intensities

Statistic	Count	Mean	Std	P25	P50	P75
Count	375.00	375.00	375.00	375.00	375.00	375.00
Mean	3,478,967.00	566.00	395.33	242.23	508.59	816.21
Std	4,177,432.00	361.75	249.54	202.85	340.19	481.52
P25	771,651.50	276.78	189.24	78.00	245.00	438.00
P50	2,287,071.00	553.24	397.95	213.00	465.00	813.00
P75	4,088,560.00	833.59	575.18	358.00	743.50	1,171.58

But we will not use the whole imaging when training the model, what if we counted only tumor area voxels intensities? Table below shows the statistics for them:

Table 7
Only tumor-area intensities

Statistic	Count	Mean	Std	P25	P50	P75
Count	375.00	375.00	375.00	375.00	375.00	375.00
Mean	19,115.23	624.70	136.13	528.70	612.75	707.98
Std	33,789.71	491.05	121.09	414.19	485.79	564.55
P25	1,657.00	225.15	44.87	192.50	222.50	254.25
P50	5,335.00	584.81	107.97	499.09	571.14	652.98
P75	21,382.00	932.36	194.93	788.00	896.00	1,046.50

Since ADC and T2 modalities are translated differently, we need to analyze their intensities seperately. Let's handle T2 first. The graph below shows the distribution of intensities of all images on average and varying percentiles:

2.4.1 T2 Intensity Analysis

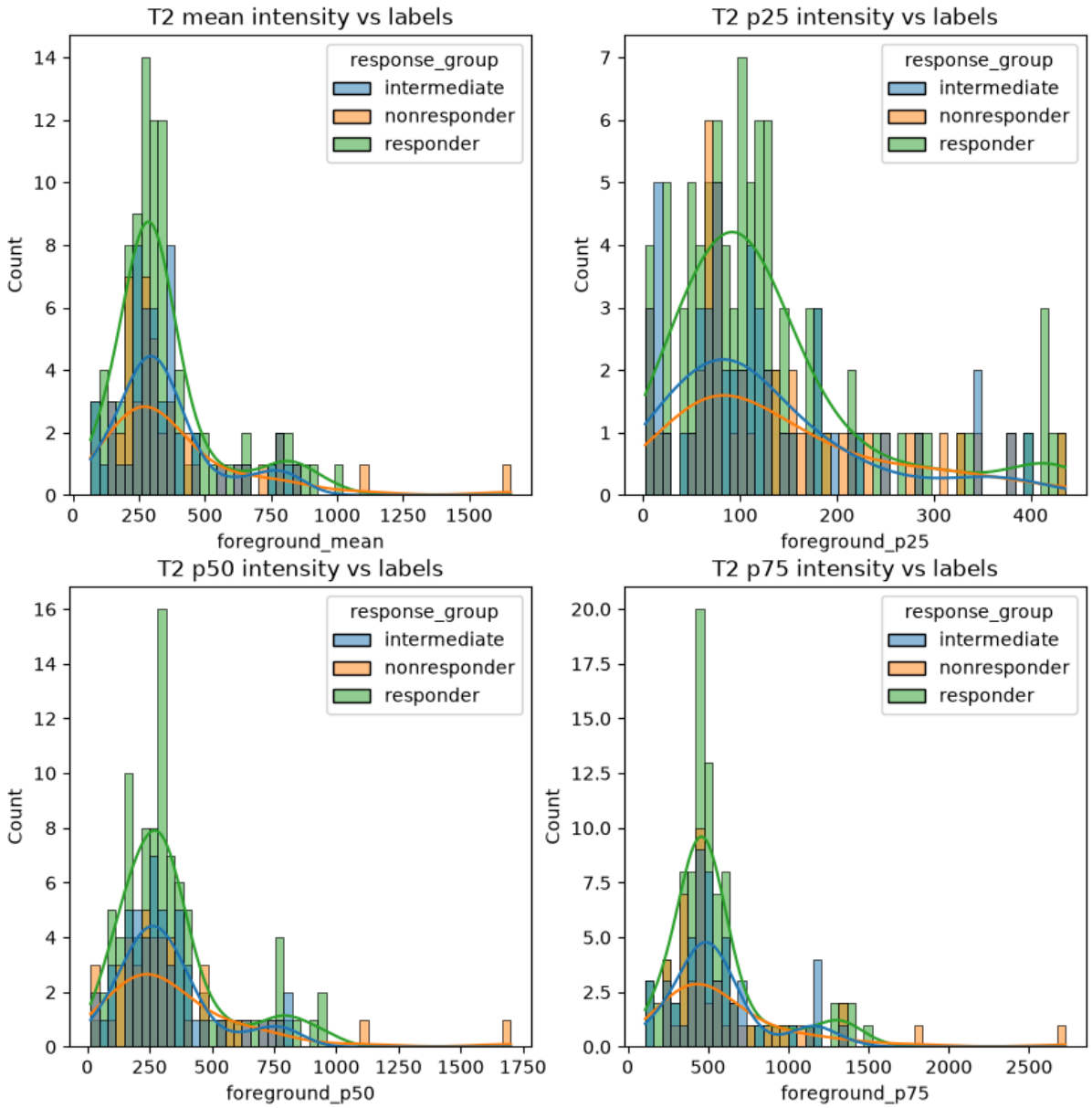


Figure 5: T2 intensity distribution

As can be seen, intensities are differing imaging to imaging a lot. Z-score normalization would fix the variation. Below you can see the same distribution after normalization:

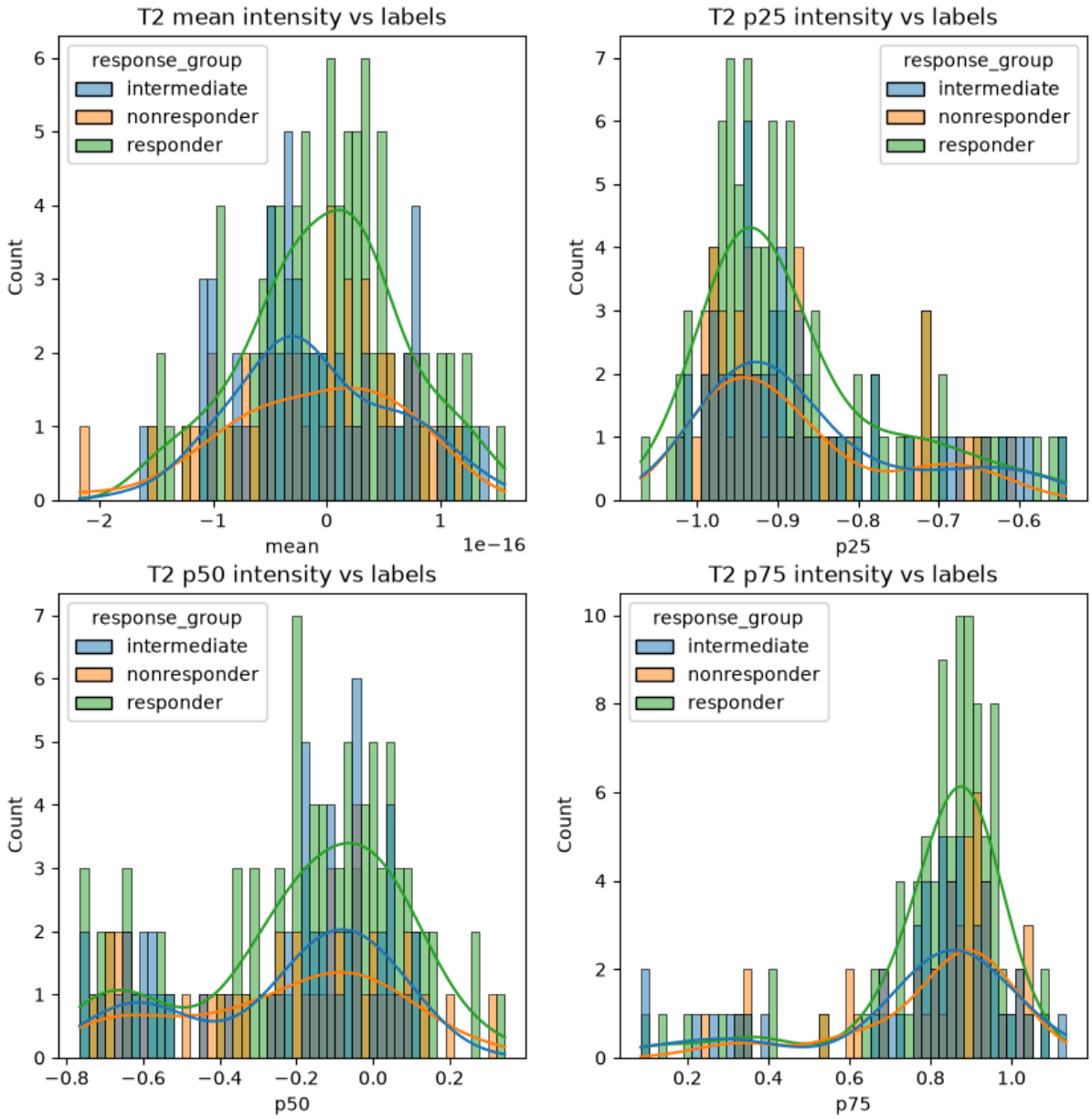


Figure 6: T2 intensity distribution normalized

Clearly, differentiation is less, absolutely resulting better in training due to elimination of noise and equalizing the domains of voxel intensities.

Any type of normalization have to be applied carefully, causing the image to lose information or increasing the noise is not acceptable. In our case, the z-score normalization we applies is a standard one, equalizing **mean=0** and **std=1**. Below you can see the difference of normalized and raw imaging format:

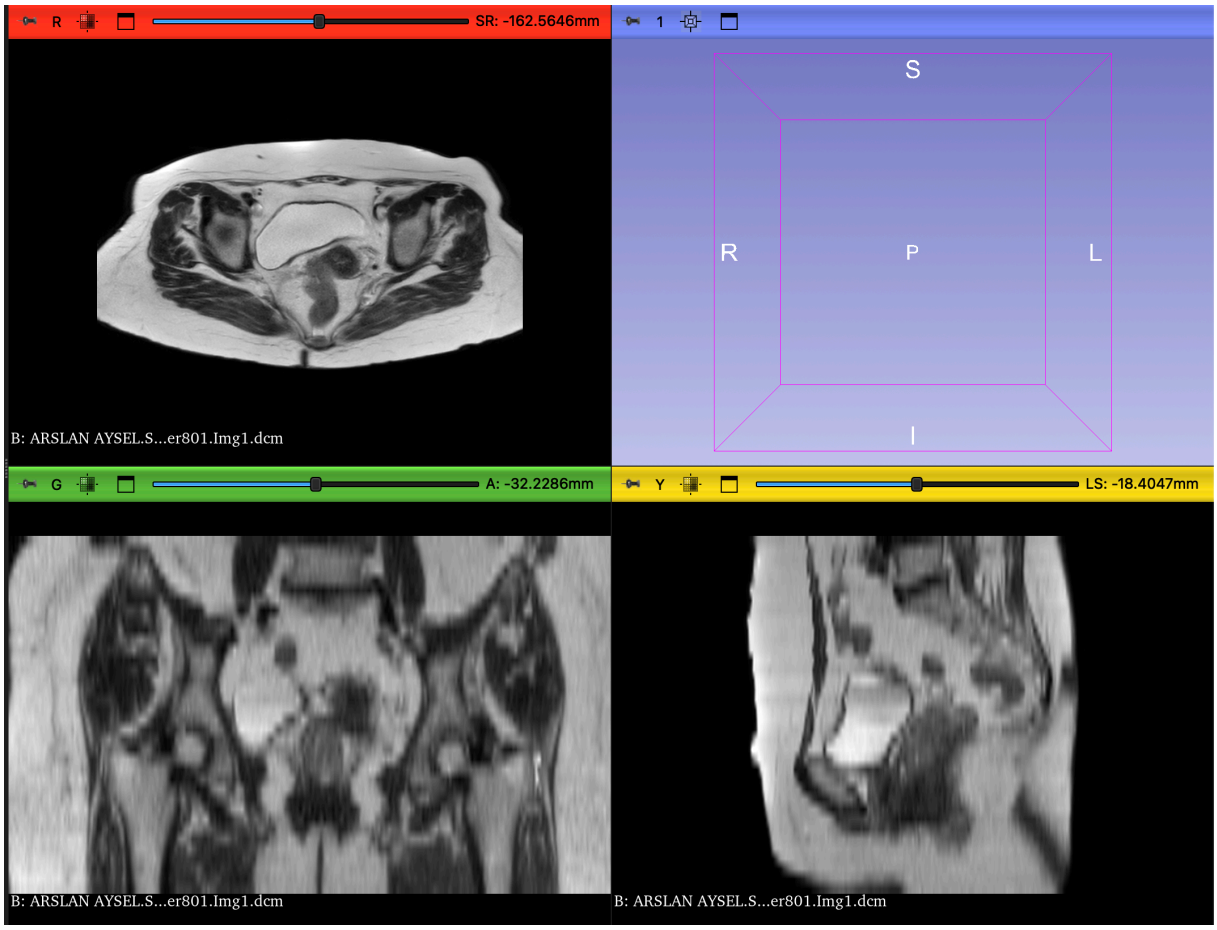


Figure 7: Original imaging

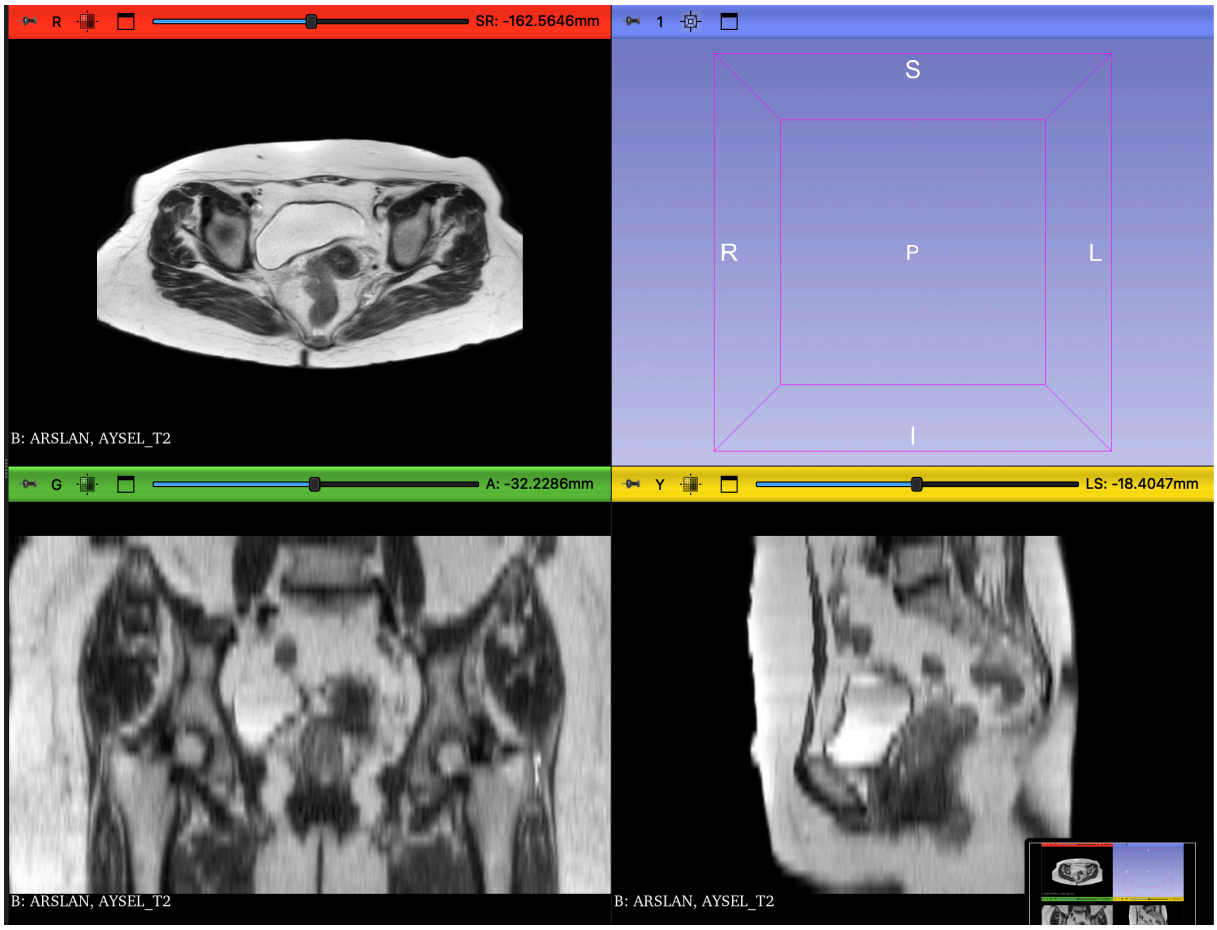


Figure 8: Z-score normalized imaging

In our case however, we applied the normalization successfully and proved it does not harm the data, so it needs to be done in the way we proposed. **[3]**

2.4.2 ADC Intensity Analysis

As we said, intensity normalization should not be applied to ADC imagings. However, still analyzing the intensity might be useful to detect some broken imagings and strengthen the insights of us over the dataset. In the graphs below, you can see the distribution of all imagings' average intensities:

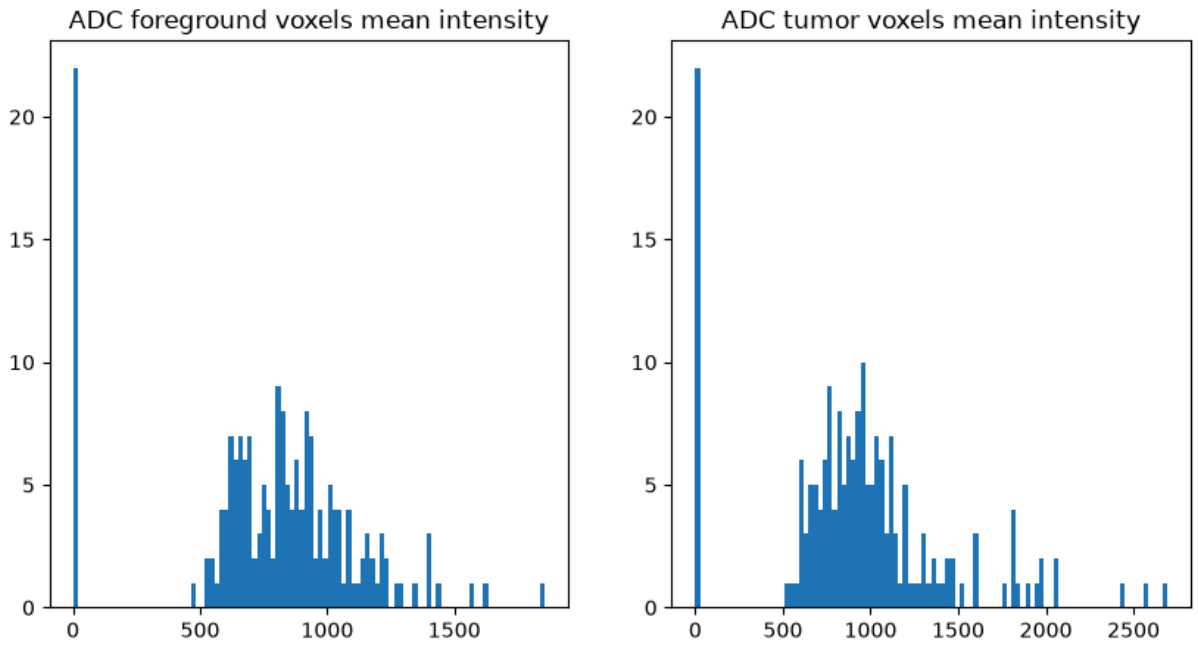


Figure 9: ADC mean-intensity distribution

Attention! There are some critical outliers among the ADC imagings. Somewhere very close to 0, there exists more than 20 imagings, which might indicate a unit difference. Machines can outputs the ADC quantities in $\frac{\mathbf{mm^2}}{\mathbf{s}}$, $\mathbf{10^{-3}}\frac{\mathbf{mm^2}}{\mathbf{s}}$ or $10^{-6}\frac{mm^2}{s}$. Unfortunately this information have not been given in the manifest, so have to figure it out ourselves.

First of all, let's figure out how many of these **small-units** samples exists:

Table 8
Distribution of ADC unit groups by batch

Batch ID	Large Units	Small Units	Total
Batch 1	88	11	99
Batch 2	77	11	88
Total	**165**	**22**	**187**

11 from each batch, shows us the cause is independent of batches. Now let's figure out the exact scale difference between large-units and small-units samples.

Table below indicates the median value of small-units samples and large-units samples:

Table 9
Median tumor intensity by ADC unit group

Units	Tumor P50
Large Units	919.000000
Small Units	0.931219

Now we have a very strong proof on 1000x rescaling might equalize the domains of different-units samples. Analyze each unit groups alone, and altogether with small-units rescaled 1000x:

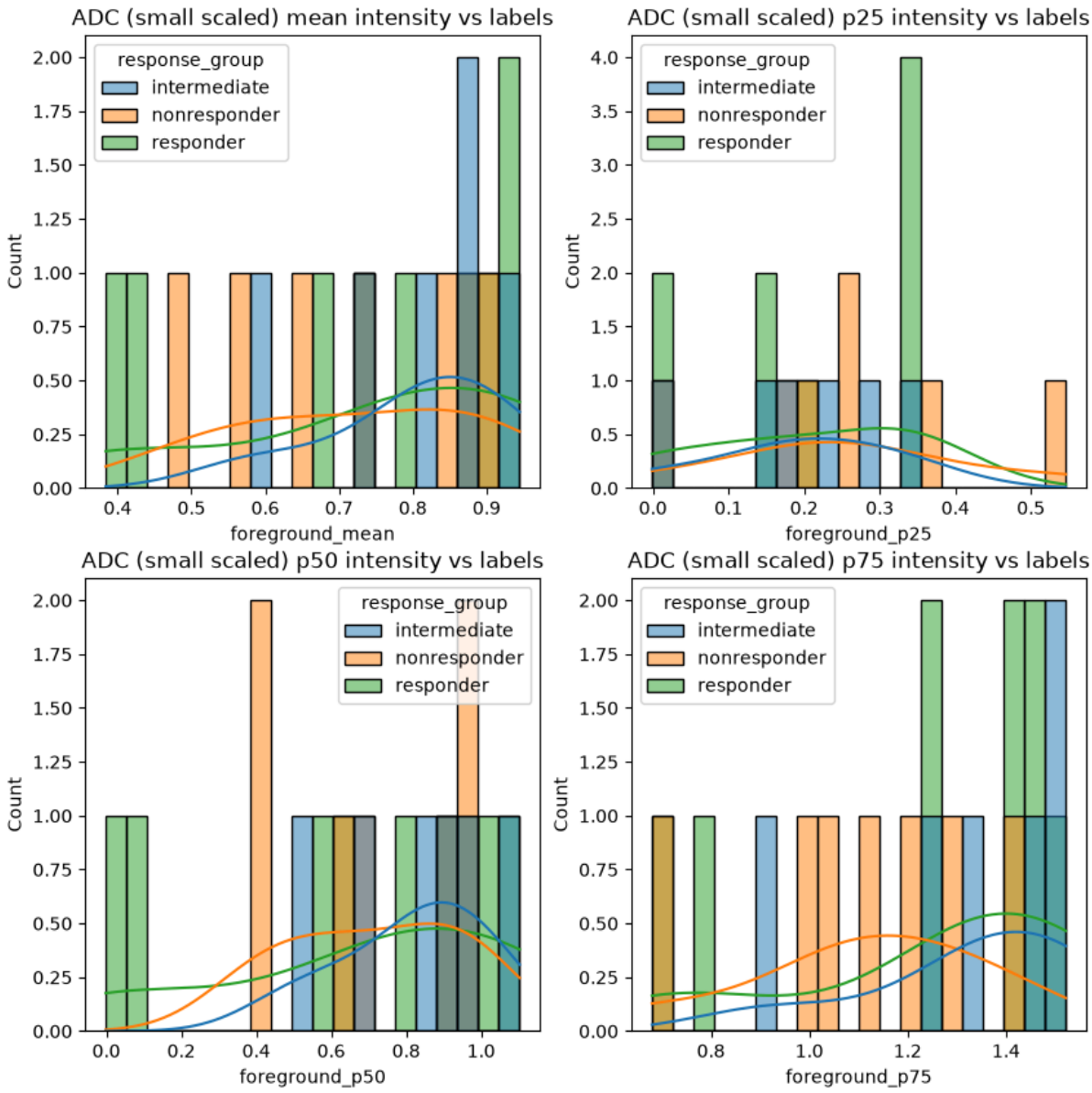


Figure 10: Only small-units

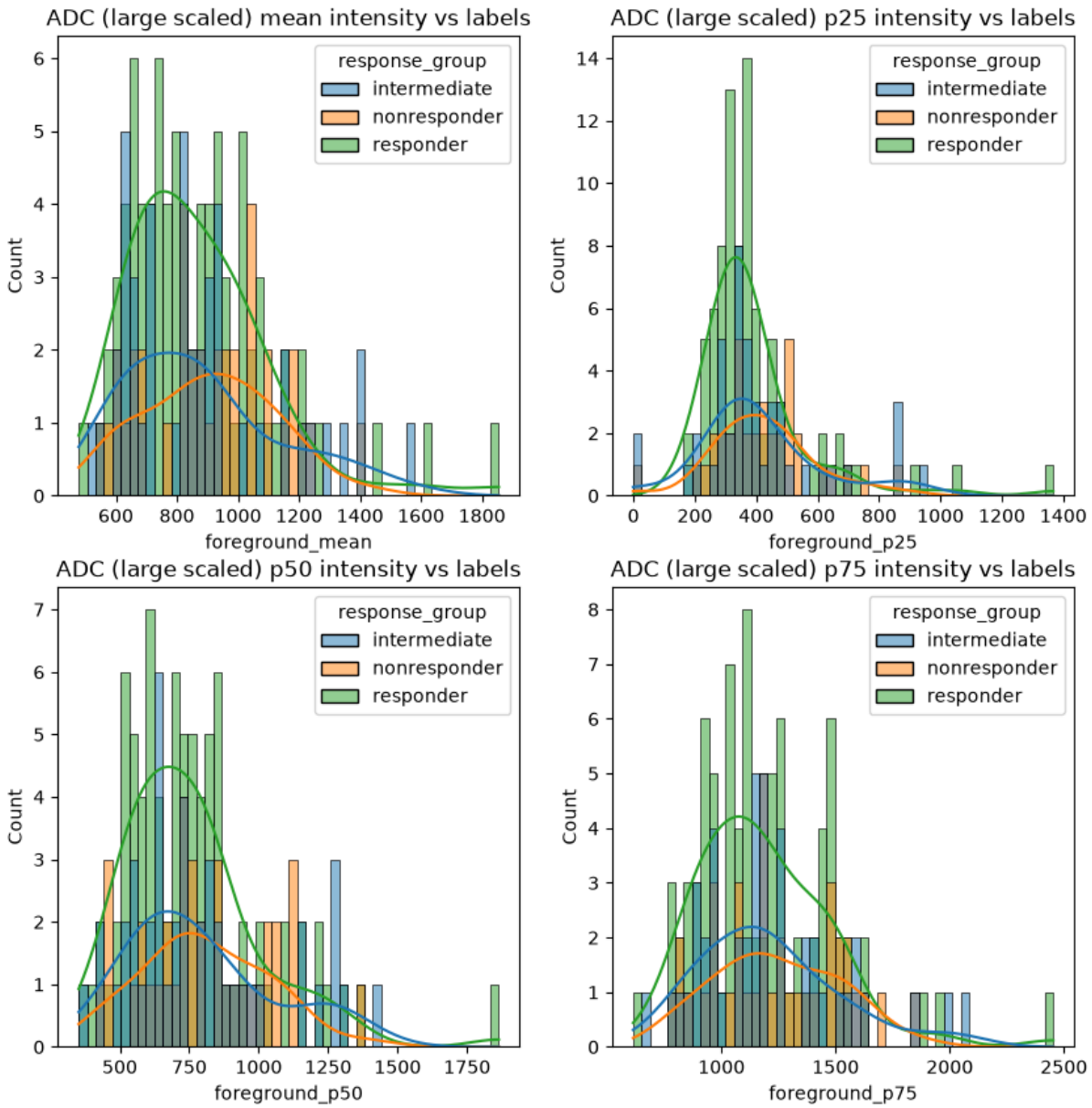


Figure 11: Only large-units

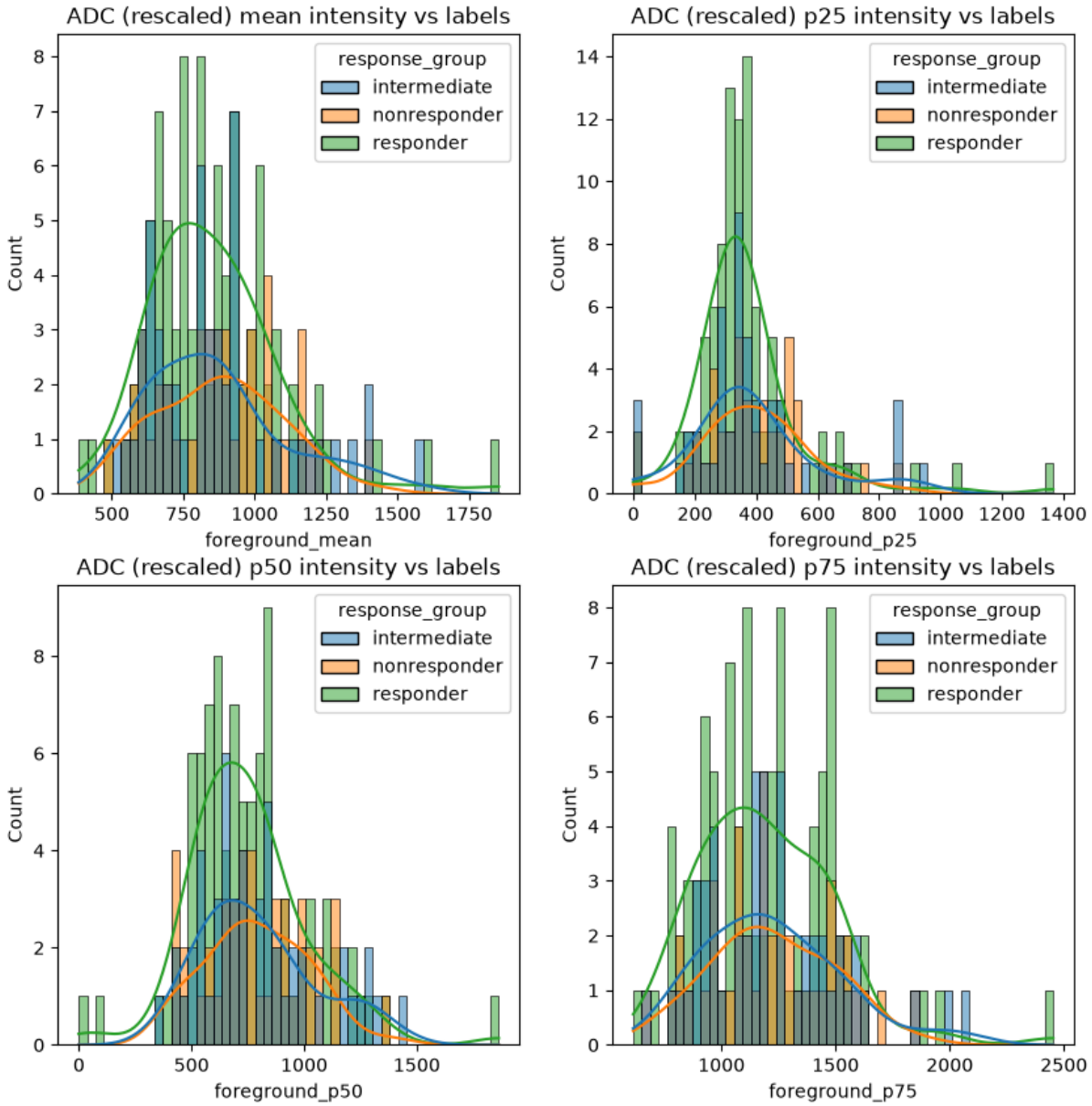


Figure 12: Altogether

Last graph shows that still there exists some values near zero, made me curious. **An ADC imaging should not have any negative values voxels, but still being near zero might mean there exists some ADC imagings with negative voxels, decreasing the average to zero.**

2.4.3 ADC Negative Intensity Analysis

This table below proves the existance of negative valued ADCs:

Table 10
ADC intensity statistics

Statistic	Count	Mean	Std	P25	P50	P75
Count	187.00	187.00	187.00	187.00	187.00	187.00
Mean	910,686.70	774.02	541.55	355.54	686.10	1,062.18
Std	480,997.00	358.08	246.80	215.89	342.54	475.87
Min	175,979.00	0.38	0.34	−1.00	−0.00	0.68
P25	633,145.00	644.08	432.97	273.50	543.00	917.00
P50	771,392.00	814.76	564.61	340.00	701.00	1,120.00
P75	1,035,282.00	973.36	690.35	443.50	854.00	1,356.50

What made me miss the existance of negative values was not to check minimum intensities in each sample. Existance of negative values in ADC is not acceptable, we need to question this further. When I query negative value containing ADC imagings **25 sample pops up:**

1. CELIK, OSMAN
2. KOZIK, TUNCAY
3. MERDIM, BURHAN
4. TASIMOVA, GULBARSHIN
5. TUZ, GULSEN
6. ANDELKOVIC, GORAN
7. OZYURT, ASIYE
8. RACHEV, NIKOLAY
9. SIS, CEMAL
10. AKBAS, EMINE MELEK
11. GOK, KADRIYE
12. SAFAROVA, ILAHA
13. BUTUN, ASUR
14. RASHIDOV, BAKHITZHAN
15. YETIMLER, REMZIYE
16. CALISKAN OZALP, BAHAR
17. OBI, NDIBE JOSEPHAT
18. ALIYEV, HASAN
19. AYATA, EMINE
20. BIGIRA, JOHN BOSCO
21. CIFTCI, LEVENT
22. ERGUNE, TELAT
23. KOCYIGIT, GULFIDAN
24. ORKUN, AYSEL
25. TEZEL, OMER

After checking randomly, I realized **negative values are backgrounds.** However, we need to prove this by checking **the weight of the negative voxels**, they have to be small numbers enough to not to change any information in the imaging, when we zero them straight:

Table 11
Mean intensity of negative-valued ADC samples

No	Patient Name	All Mean	Neg Mean
1	CELIK, OSMAN	0.8622	−0.0229
2	KOZIK, TUNCAY	668.8952	−1.0000
3	MERDIM, BURHAN	0.9406	−0.0396
4	TASIMOVA, GULBARSHIN	0.7454	−0.0352
5	TUZ, GULSEN	0.5869	−0.1223
6	ANDELKOVIC, GORAN	618.0167	−1.0000
7	OZYURT, ASIYE	0.7225	−0.1749
8	RACHEV, NIKOLAY	0.8636	−0.0311
9	SIS, CEMAL	0.8568	−0.1392
10	AKBAS, EMINE MELEK	271.6863	−256.0000
11	GOK, KADRIYE	0.4269	−0.0699
12	SAFAROVA, ILAHA	0.6851	−0.1342
13	BUTUN, ASUR	604.7395	−1.0000
14	RASHIDOV, BAKHITZHAN	0.8232	−0.0873
15	YETIMLER, REMZIYE	0.8658	−0.0109
16	CALISKAN OZALP, BAHAR	0.5775	−0.0102
17	OBI, NDIBE JOSEPHAT	0.6379	−0.0343
18	ALIYEV, HASAN	0.8021	−0.0490
19	AYATA, EMINE	0.3839	−0.0625
20	BIGIRA, JOHN BOSCO	0.9102	−0.0498
21	CIFTCI, LEVENT	0.9446	−0.0432
22	ERGUNE, TELAT	270.0076	−256.0000
23	KOCYIGIT, GULFIDAN	0.8754	−0.0654
24	ORKUN, AYSEL	0.7288	−0.0862
25	TEZEL, OMER	0.9212	−0.0500

Caution: Two imagins has **incredibly high negative means**, −256. Fortunately, I checked them manually on 3d Slicer and they are just background. Can be zeroed safely.

My conclusion from here is, just **zero the negatives straight**.

The altogether distribution after zeroing all negatives:

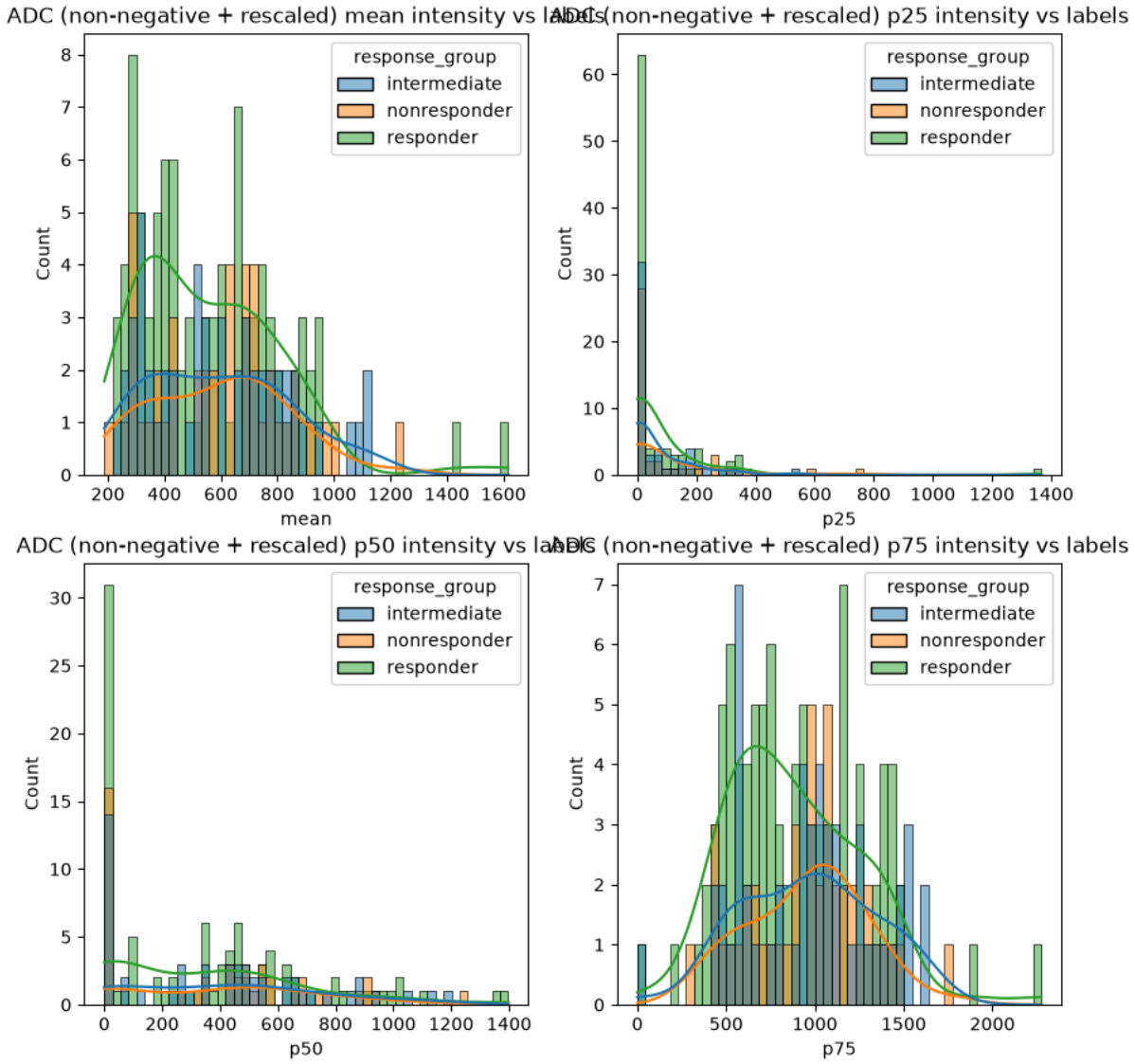


Figure 13: Non-negative rescaled ADC intensity analysis

From mean graph, we can see all imagings are distributed naturally. In other graphs many zeros might seem dangerous but they are not, because they are percentiles and thanks to how big the background is, all most of the imagins have large zero areas all over their surfaces. **After random manual check on 3D Slicer, I approve the images are fine.**

Comparison of an image before/after manipulation:

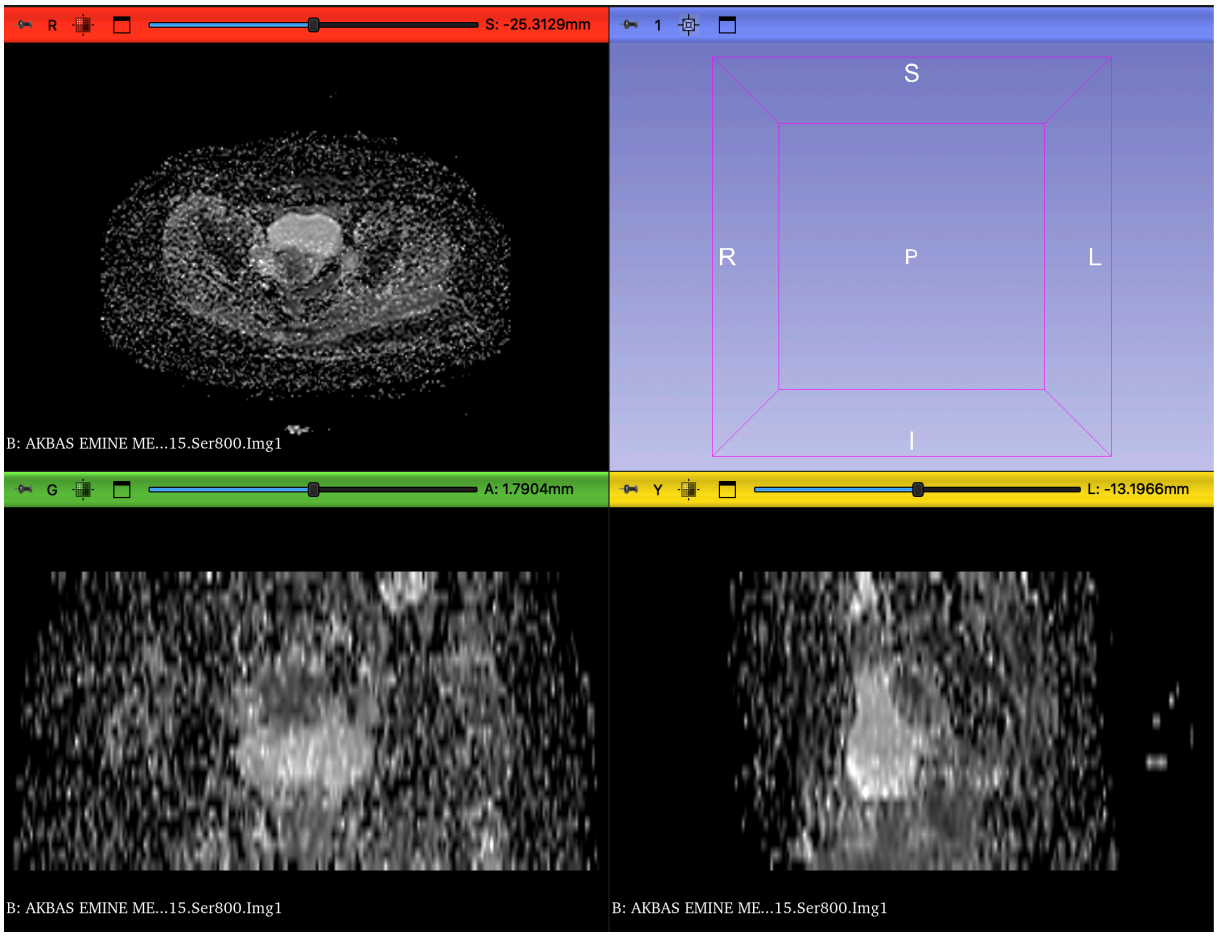


Figure 14: ADC before manipulation

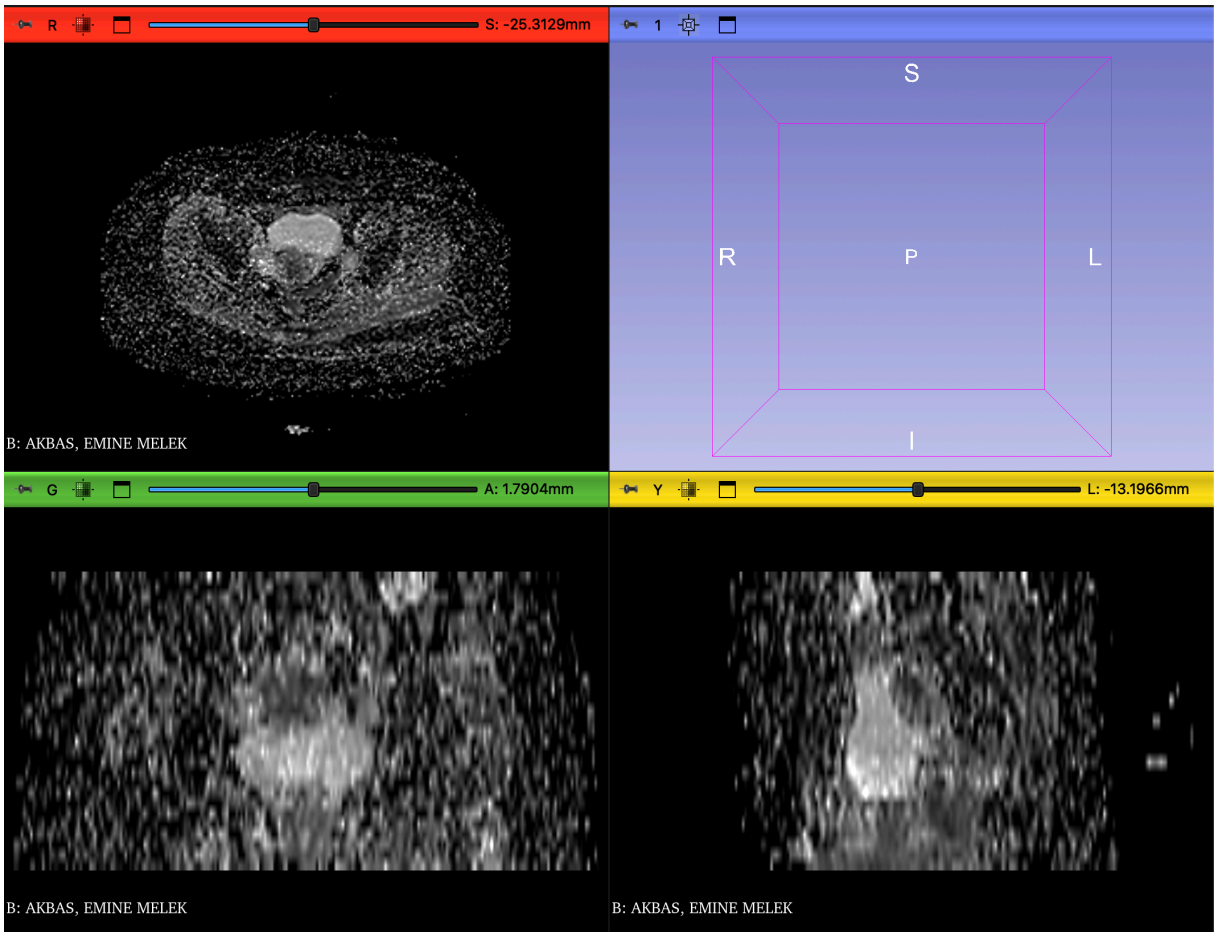


Figure 15: ADC after manipulation

So zeroing negative values and rescaling small-unit ADC imagings can safely be applied to the data. **[4]**

3 Roadmap to Preprocessing

After all the analysis, we will be applying all the manipulations [1], [2], [3] and [4]:

1. Eliminate patient GUL, ABDULLAH.
2. Crop the tumor area and resample the shape and spacings.
3. Apply z-score intensity normalization on T2 imagings.
4. Zero negative values and rescale x1000 small-unit ADC imagings.

Then, dataset should be ready to training. However, training process might give us idea about the relevance of the preprocessing.

Additionally, this one is not for preprocessing but, **don't forget the [!], use class weighted loss and statified train-test seperation when training.**

4 Conclusion

When all the dataset is carefully analyzed, it's clear that the dataset is **far from being a clear dataset**, in order to achieve a good performance with the trained model, not only training process should be applied incredibly carefully, but also exploratory data analysis and preprocessing has to be applied overly cautious. Otherwise, data might lose information, or domain difference between samples on intensity, spatial aspects or any other aspects, training won't be proper and bad results will be

unavoidable. **However, even all the exploratory data analysis, preprocessing and training done perfectly, still any meaningful results might not be achieved**, because **(1)** dataset has incredible imbalances and domain difference in many aspects in it's raw form and **(2)** number of samples might not be enough for a Q1 study. Nevertheless, we should try our bests. Still all the work done so far and in the future might be an infrastructure even not any satisfying results achieved, but we might be able to find better and more samples and refine the work.

Stay sharp and smart!